

Tushar Barua

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Summary

Full Stack Web Developer specialized in Next.js, React, TypeScript, and Node.js with a strong foundation in Data Structures & Algorithms (DSA). Experienced in architecting production web applications serving thousands of users. Passionate about building scalable, high-availability systems and privacy-preserving healthcare solutions.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C/C++, SQL
Frontend: Next.js, React, Tailwind CSS, Framer Motion, HTML/CSS
Backend & Databases: Node.js, Express, Prisma ORM, PostgreSQL, MySQL, REST APIs
ML & Research: Federated Learning, Explainable AI (XAI), Scikit-learn, Pandas, Flask
Tools & OS: Git, GitHub Actions, Docker, Linux (Pop!_OS, WSL), Postman, Figma
Concepts: System Design, Scalability, High Availability, Database Optimization, OOP

Experience

Full Stack Developer (Independent Contractor) 2022 – Present
Self-Employed *Remote*
– Architected 3+ scalable, high-availability web applications end-to-end from system design to CI/CD automated deployment.
– Designed optimized distributed data models using PostgreSQL and Prisma ORM, ensuring data consistency and sub-second queries.
– Built MediScript-E, a healthcare platform digitizing workflows for 300+ patients and reducing appointment scheduling by 70%.
– Developed an interactive DSA Platform serving 500+ active developers, improving user problem-solving success metrics by 65%.
– Integrated Machine Learning models into production web applications via robust Python-based REST APIs.

Research & Thesis

Fed-XAI: Privacy-Preserving Health Risk Prediction Framework 2024 – Present
– Co-developed a decentralized framework combining Federated Learning and XAI to monitor occupational health risks in Bangladesh.
– Designed communication models ensuring zero data leakage while maintaining a 92% model evaluation accuracy.
– Presented core abstract concepts and empirical research methodologies at a poster presentation event hosted by Chittagong University.

Projects

MediScript-E | *Next.js, Node.js, Prisma ORM, PostgreSQL, Tailwind CSS* 2024 – Present
– Live: mediscript-e.vercel.app GitHub: github.com/tusharsno/mediscript
– Secure digital healthcare system supporting e-prescriptions, flexible appointment slots, and automated record tracking.
– Engineered a robust role-based access control (RBAC) layer for patient-doctor spaces with enhanced privacy controls.
DSA Platform | *Next.js, TypeScript, Tailwind CSS, PostgreSQL* 2024 – Present
– Live: dsaplatform.vercel.app GitHub: github.com/tusharsno/dsa-platform
– An interactive software platform facilitating algorithm mastery and progress tracking with dynamic analysis visualizations.
Heart Disease Predictor | *Python, Scikit-learn, Flask, Next.js* 2023
– Live: heart-scan-three.vercel.app GitHub: github.com/tusharsno/heart-scan
– Designed an early heart hazard detection ML pipeline securing 92% accuracy with extremely responsive inference times.

Education

B.Sc. in Computer Science and Engineering (Ongoing) 2023 – Present
University of Science and Technology Chittagong (USTC) *Chittagong, Bangladesh*
– **CGPA: 3.7+ / 4.00**
– Relevant Coursework: Data Structures & Algorithms, System Design, OOP, Database Management Systems, Software Architecture.
– Collaborating with university research groups on biomedical engineering frameworks and computational modeling projects.

Competitive Programming, Leadership & Content Creation

Competitive Programming (Codeforces & LeetCode)
– Active problem solver on Codeforces (Handle: tusharsno) with 250+ solutions; polishing core skills via LeetCode.
Assistant Treasurer – Robotics Club, USTC 2022 – 2023
– Supervised club accounts, tactical budgeting allocations, and strategic fund dispersion for national tech challenges.
Programming Content Creator – The Tech Heaven (YouTube) 2023 – Present
– Publishing technical tutorials on full-stack development methodologies and DSA paradigms for aspiring Bengali developers.